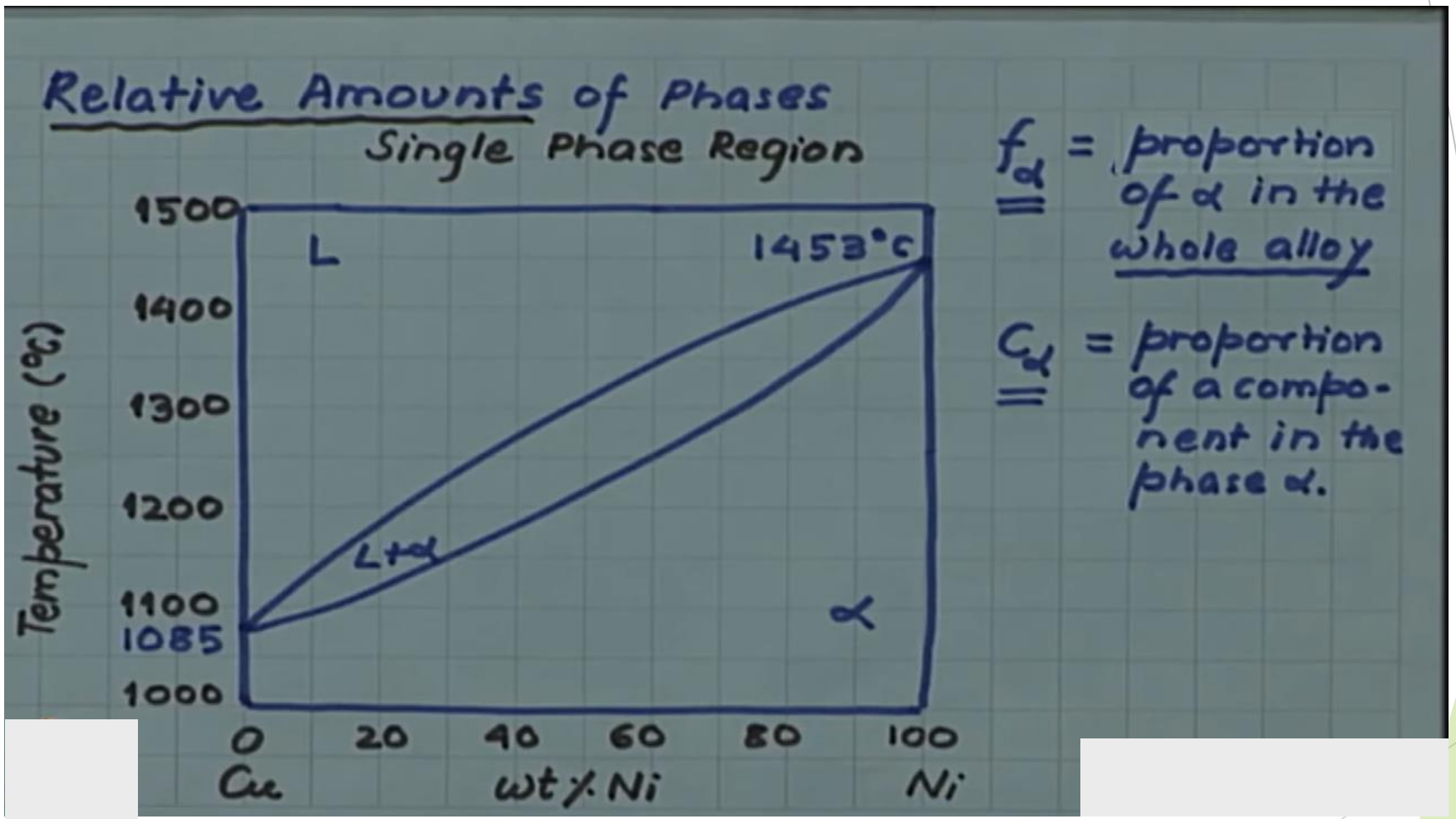
LASC MED201

Class 6

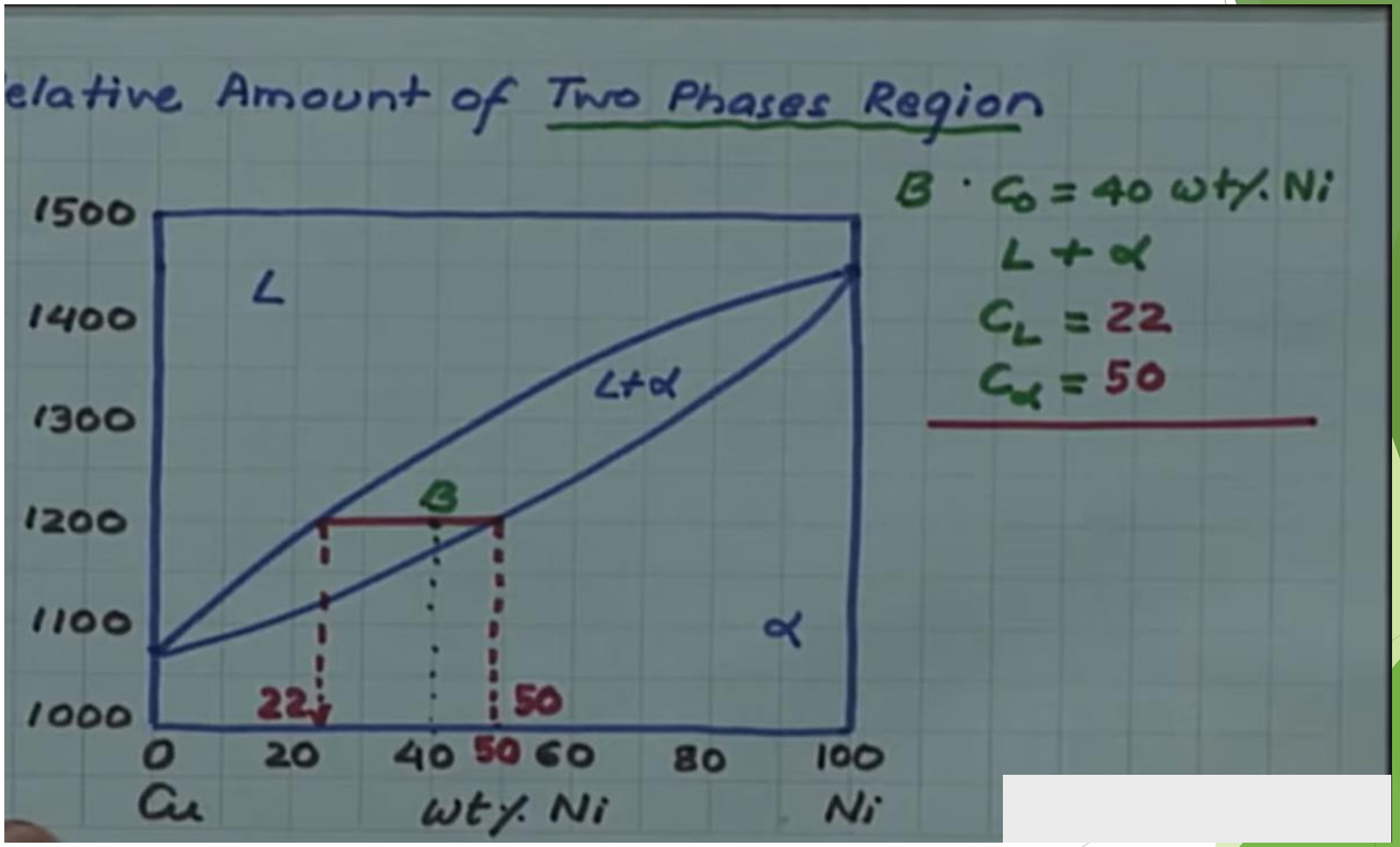
Phase Diagrams



f_α = proportion of α in the whole alloy

C_α = proportion of a component in the phase α.

Weight or Proportion are same thing. Try to follow sir's convention by using W instead of f

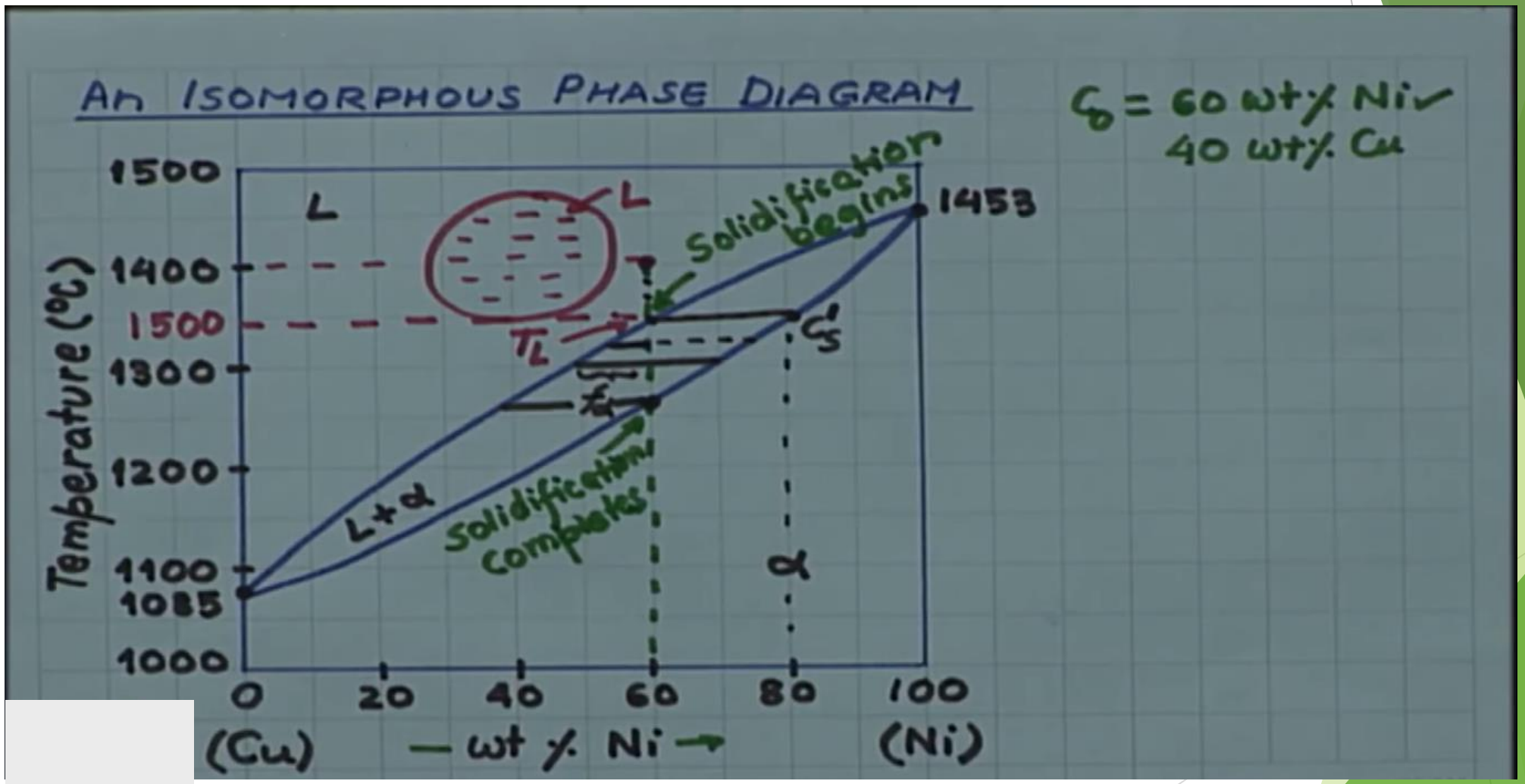


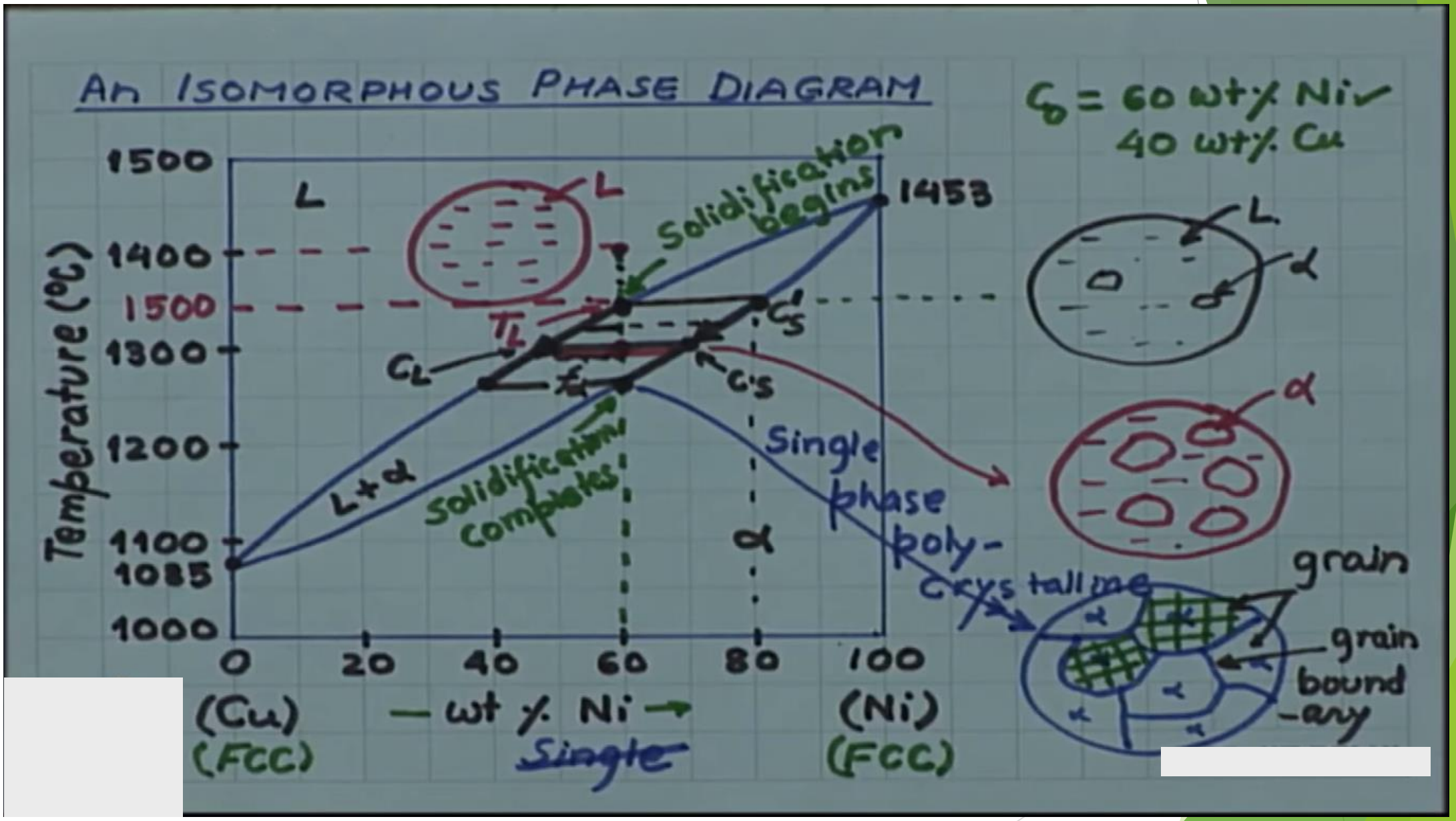
What is weight percentage of liquid and alpha phase in previous diagram?

Lever rule for sir's slide

- Answer: WL= 0.36 and Walpha=1-0.36=0.64

Cooling of Cu-Ni Alloy




An Isomorphous Phase Diagram
C_0 = 60 wt% Ni
40 wt% Cu
Temperature (°C)
1500
1400
1500
1300
1200
1100
1085
1000
L
L
Solidification begins
1453
T_L
C_L
C_S
C_S
L+α
Solidification completes
α
Single phase poly-crystalline
α
L
α
α
grain
grain bound-ary
0
20
40
60
80
100
(Cu)
(FCC)
— wt % Ni →
(Ni)
(FCC)
Single

Non Equilibrium Solidification (causes segregation)

- Very Imp ques of Cored vs eqbm structure difference.
- Ni has high mp. Forms center of core
- Cu which has low mp forms outer boundary/grain boundary
- Coring can be eliminated by homogenous heat treatment
- Cored structure due to fast rate of cooling whereas eqbm structure due to slow rate of cooling.

Eutectic System

- A **eutectic** or **eutectic mixture** is a mixture of two or more phases at a composition that has the lowest melting point.
- It is where the phases simultaneously crystallize from molten solution. The proper ratios of phases to obtain a eutectic is identified by the eutectic point on a binary phase diagram.
- Not all binary system alloys have a eutectic point: those that form a solid solution at all concentrations, such as the gold-silver system, have no eutectic. An alloy system that has a eutectic is often referred to as a eutectic system, or eutectic alloy

Reference: Prof Grewal's Slide

The temperature that corresponds to this point is known as the eutectic temperature. Solid products of a eutectic transformation can often be identified by their lamellar structure, as opposed to the dendritic structures commonly seen in non-eutectic solidification.

Pb-Sn Solder Alloy

Soldering:

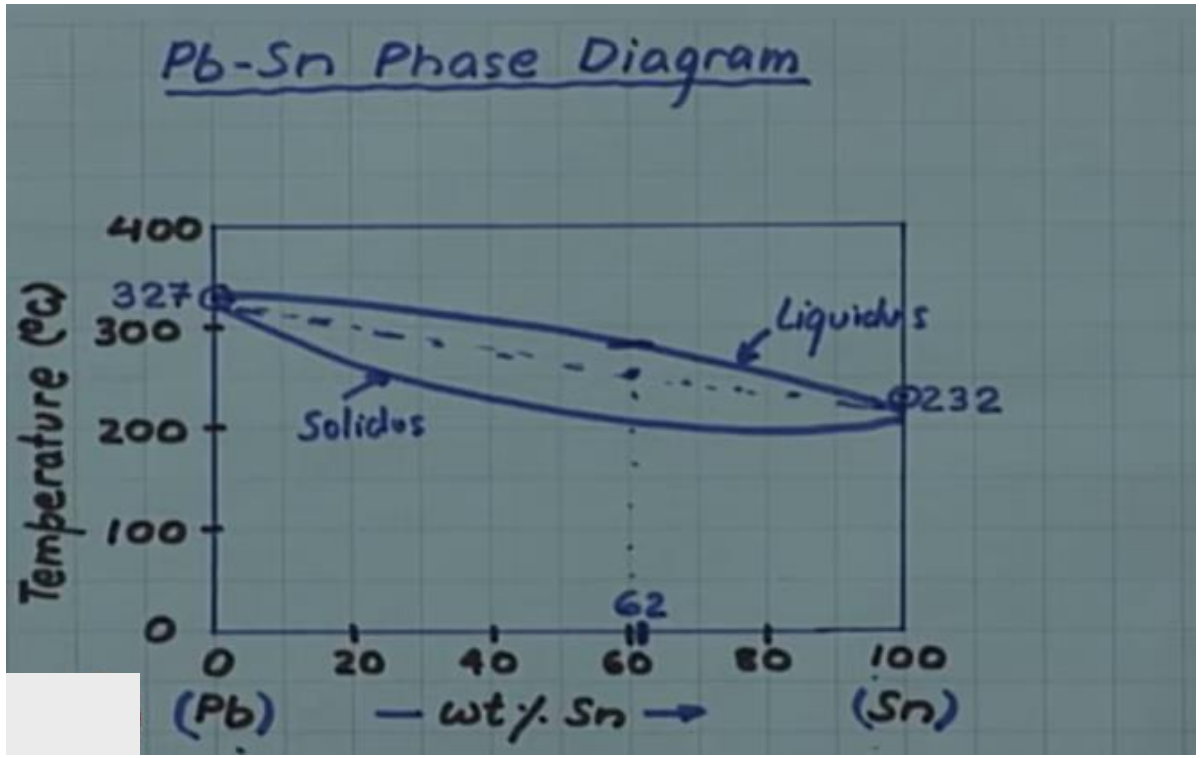
Need: Low melting alloy

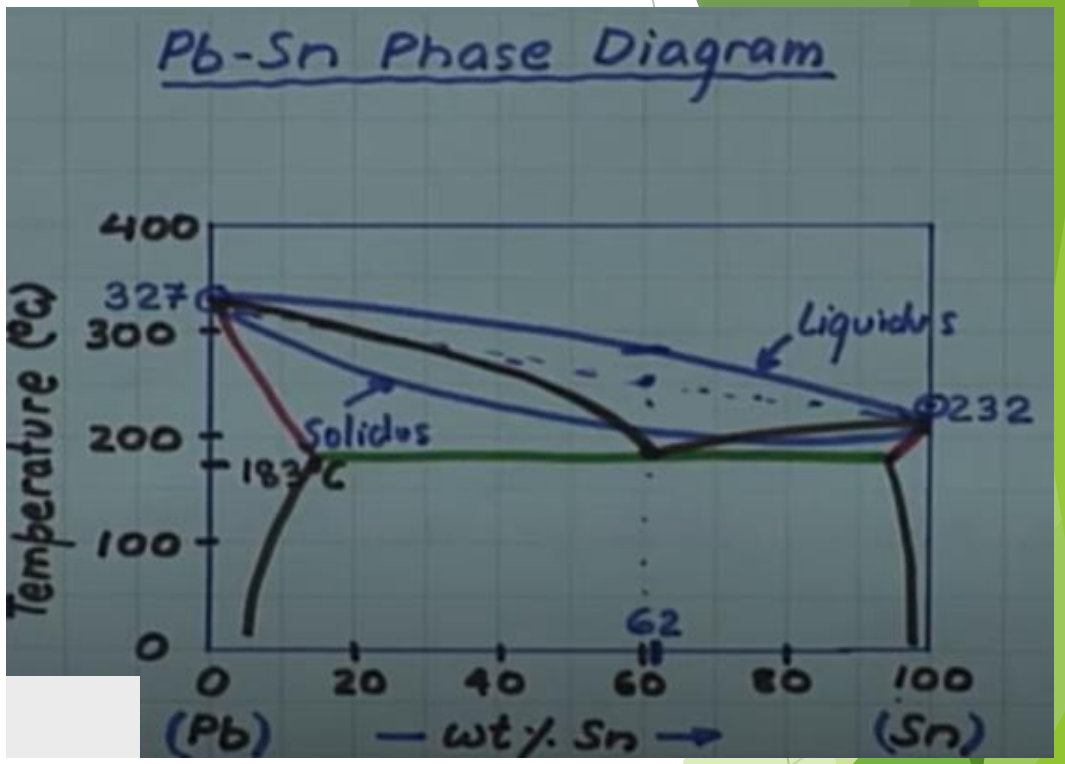
Strength (Alloy stronger than pure elements)

Pb $T_m^{Pb} = 327°C$

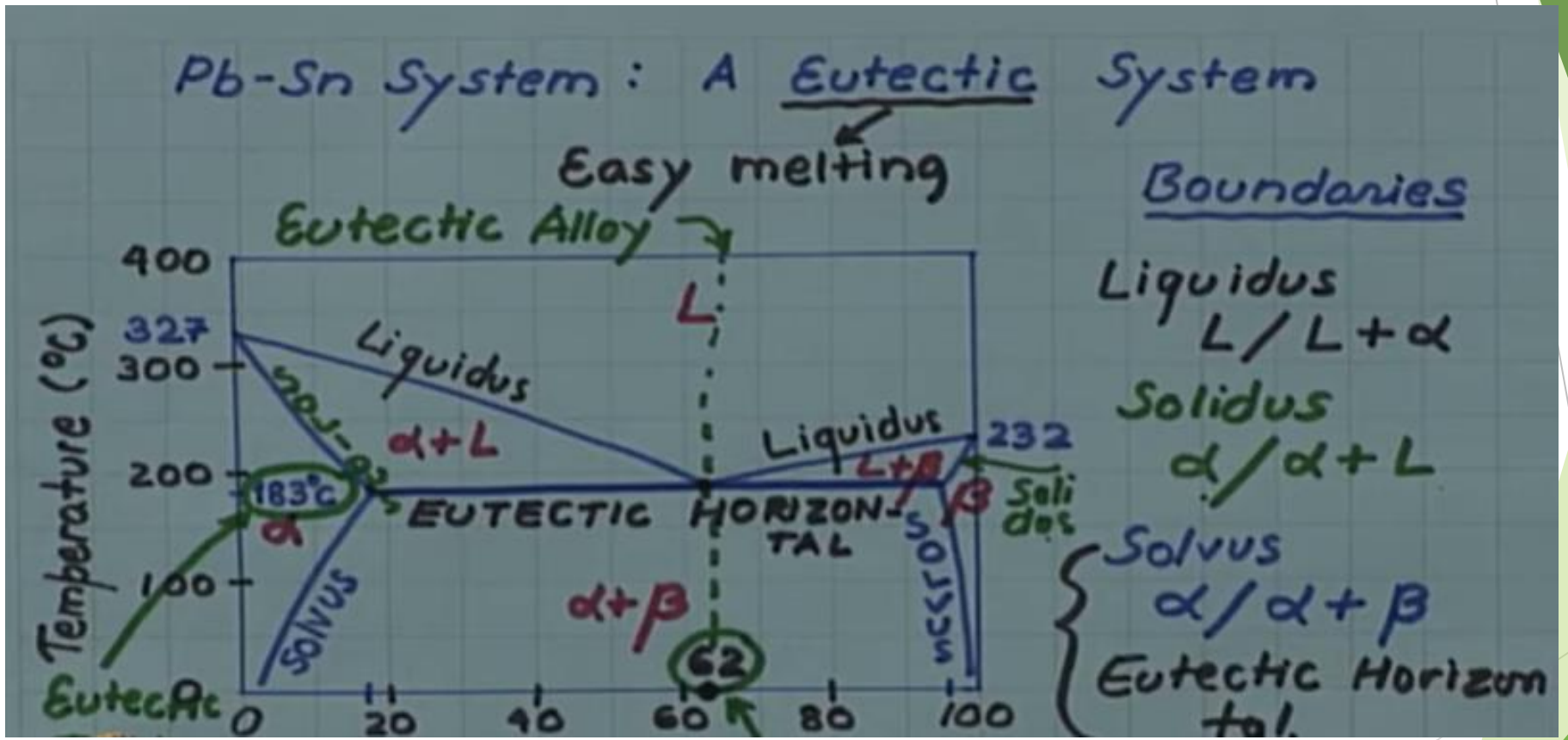
Sn $T_m^{Sn} = 232°C$

How does Eutectic point come?



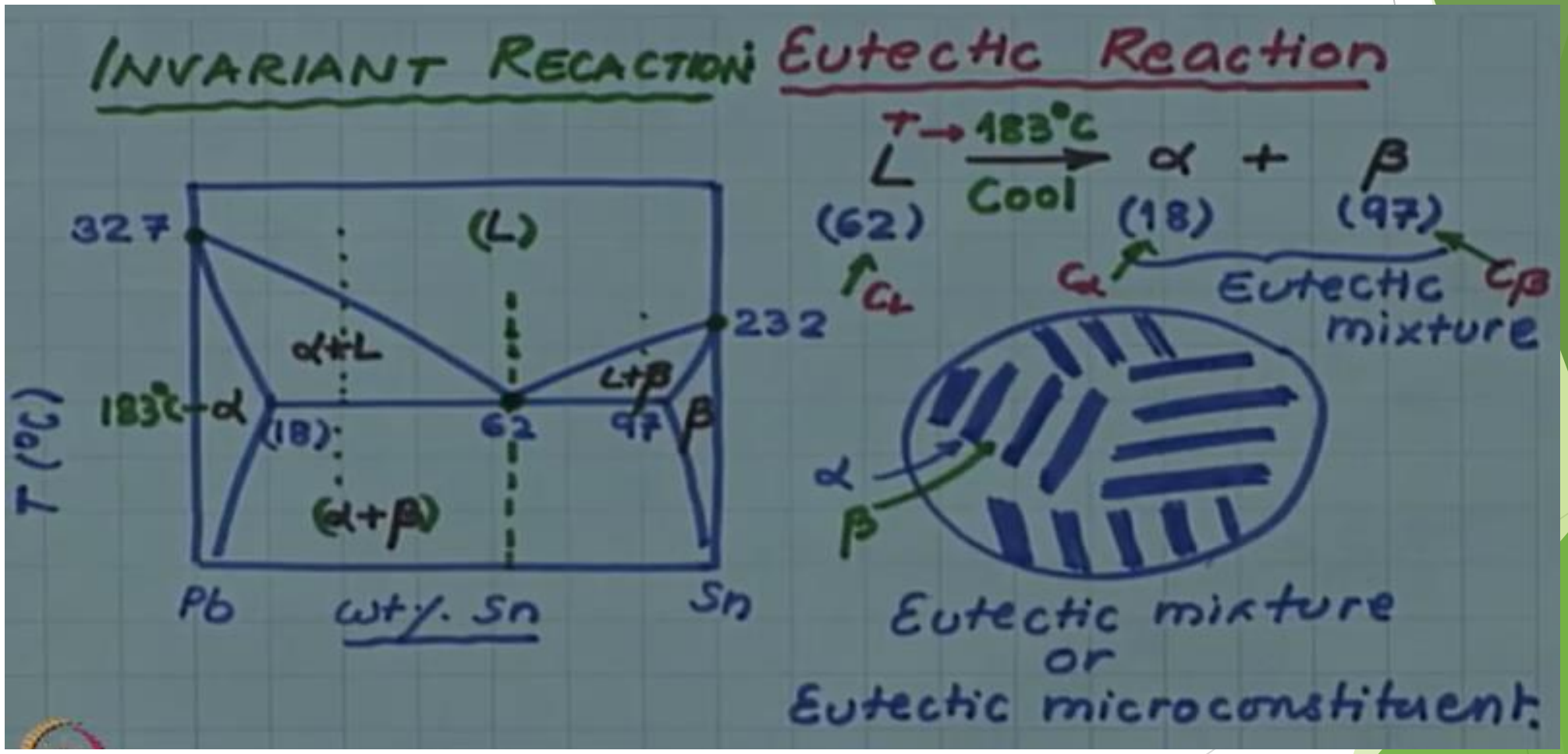


Interpretation of Eutectic Phase Diagram



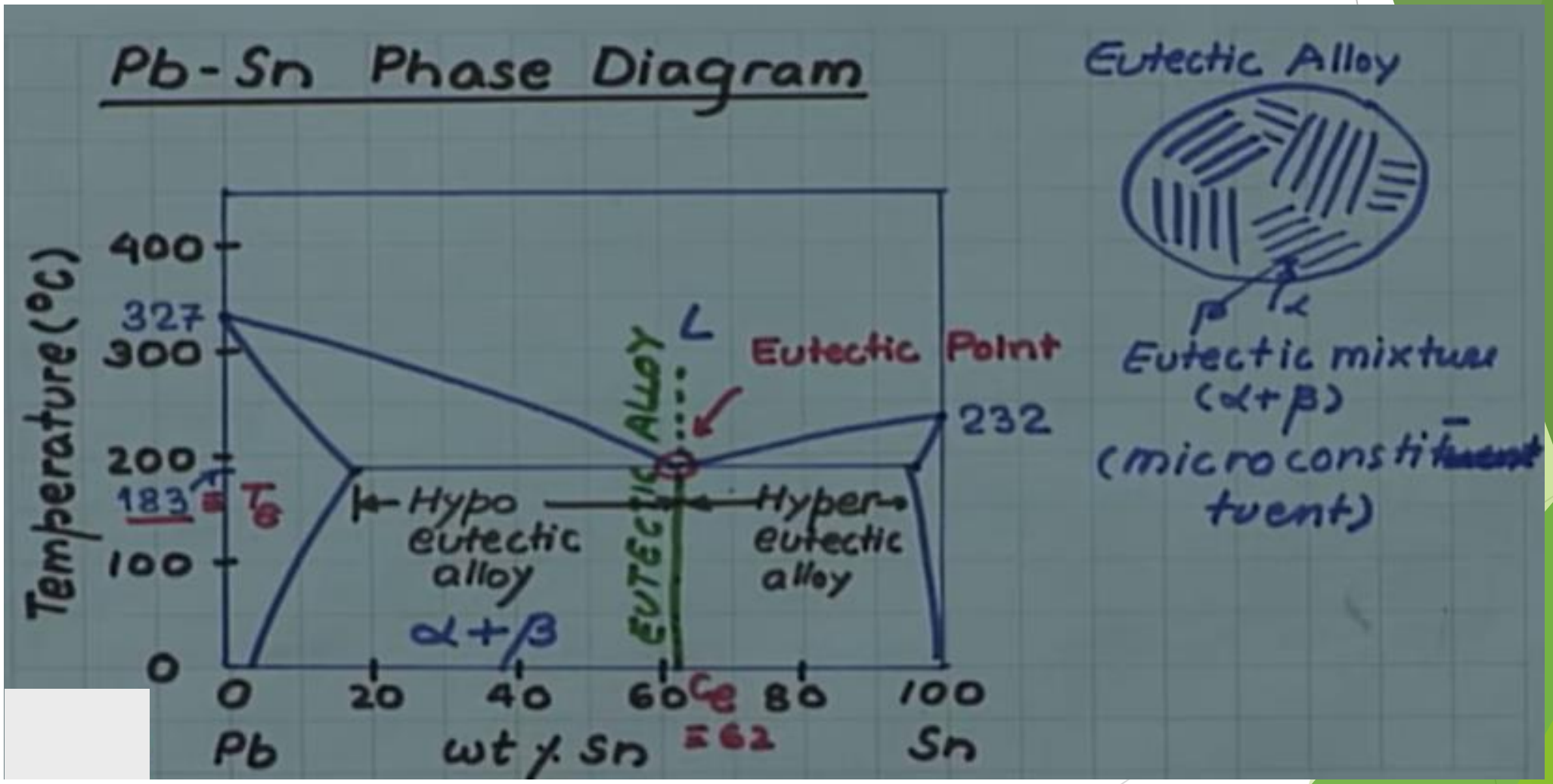
What is eutectic Composition?

Eutectic Reaction

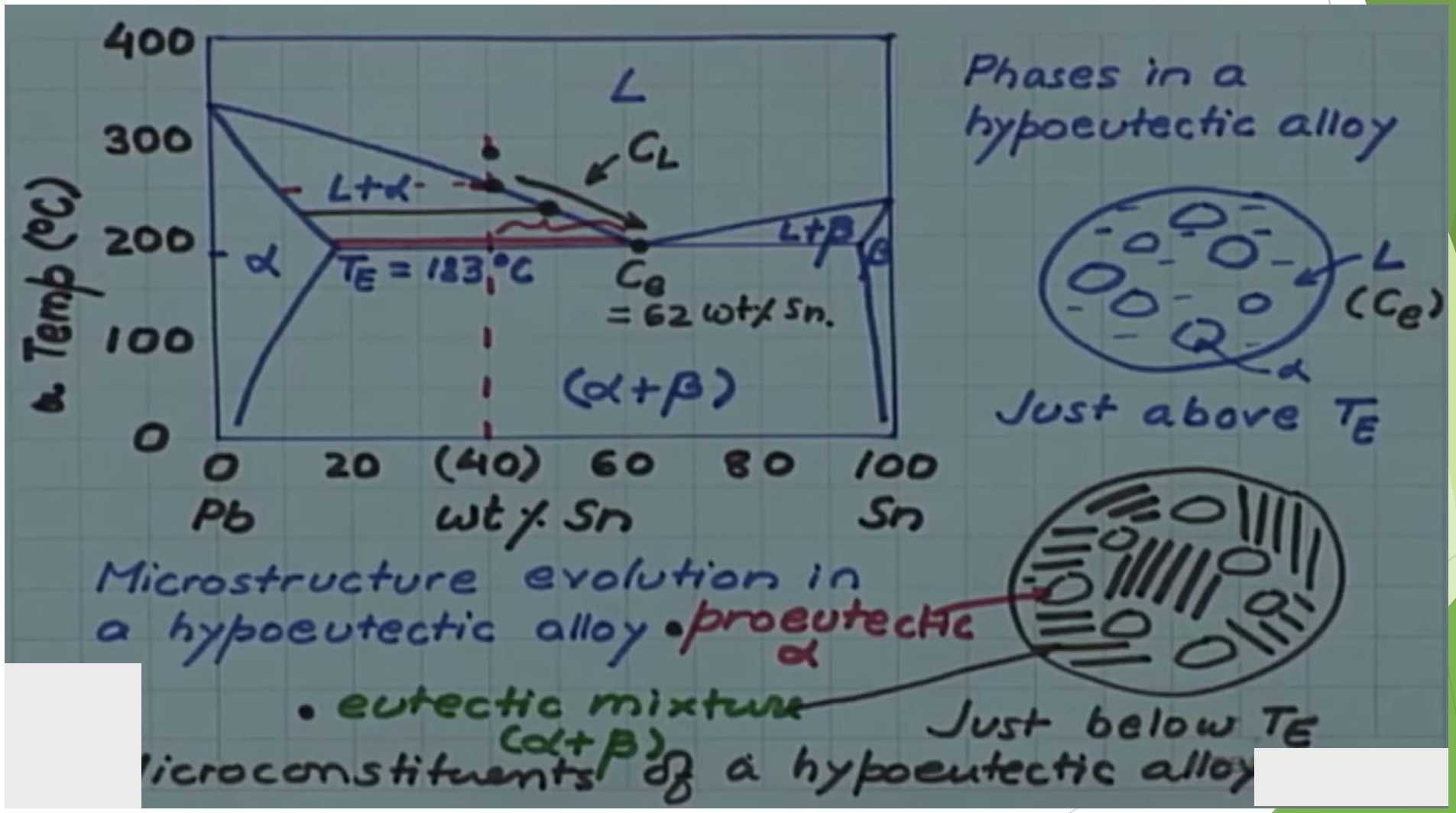


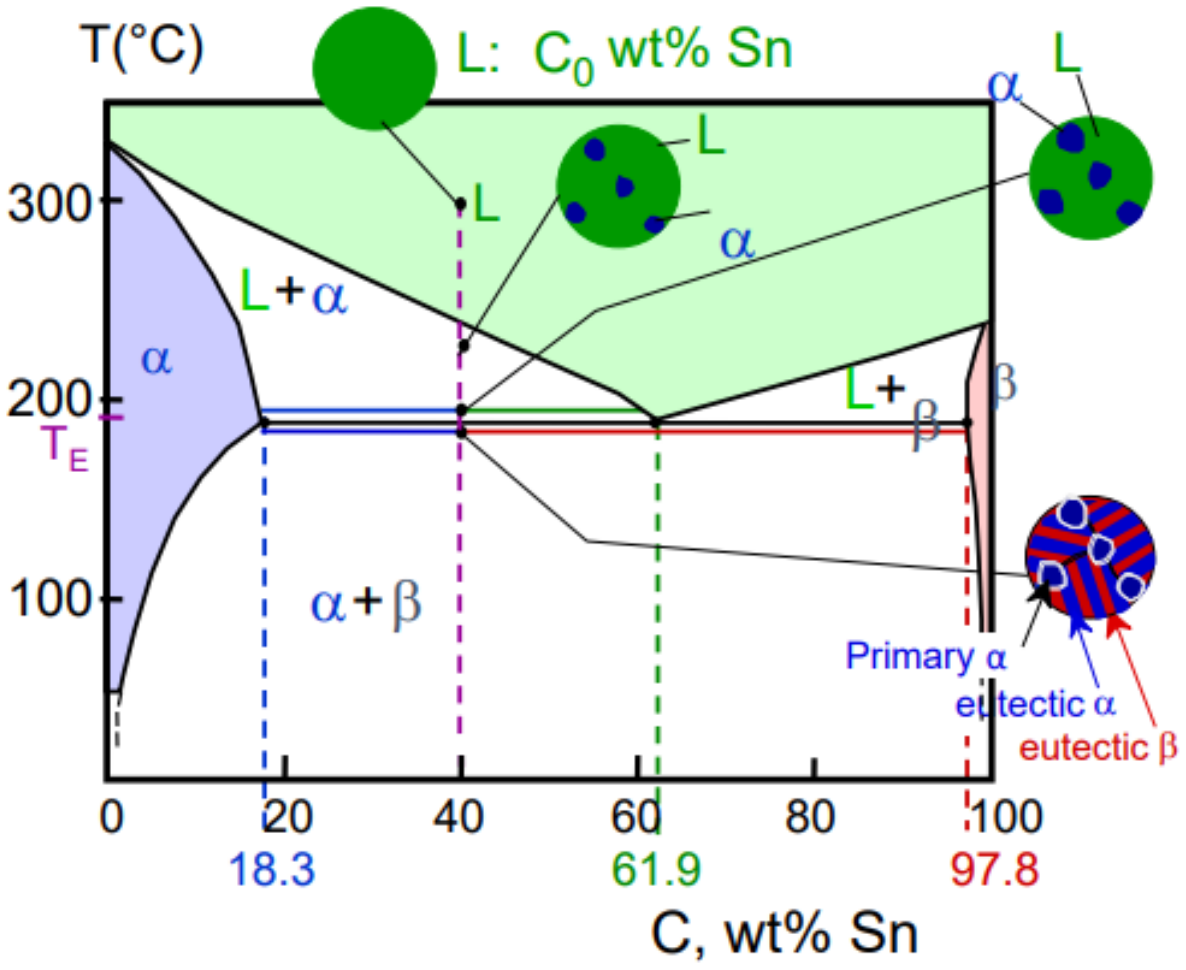
Numericals from Sir's slide

Hypoeutectic and Hypereutectic



Hypoeutectic





Calculate WL and Walpha primary?

- Just above T_E :

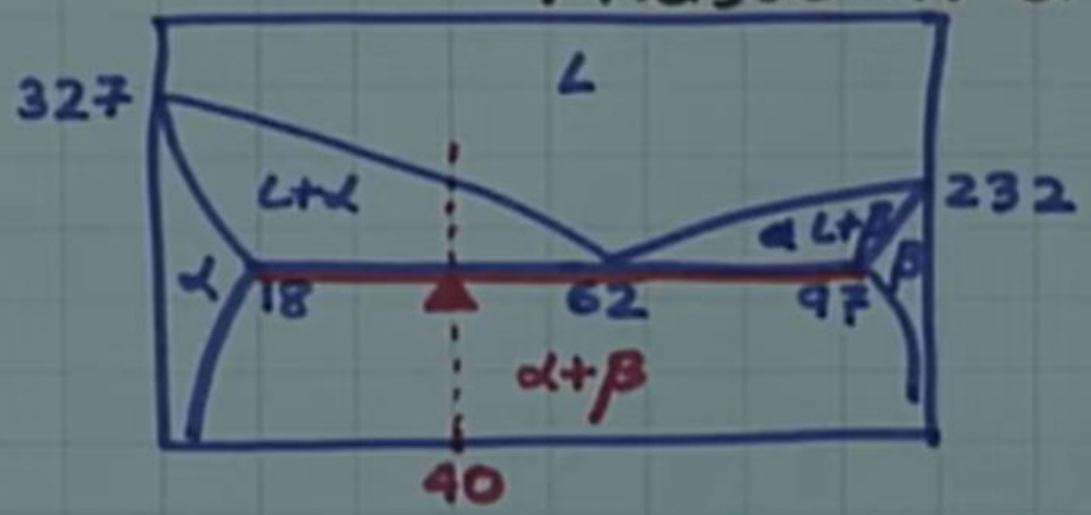
C_α = 18.3 wt% Sn

C_L = 61.9 wt% Sn

$$W_{\alpha(\text{Primary})} = \frac{C_L - C_0}{C_L - C_\alpha} = 0.5$$

$$W_L = (1 - W_\alpha) = 0.5$$

Phases in a Hypoeutectic Alloy



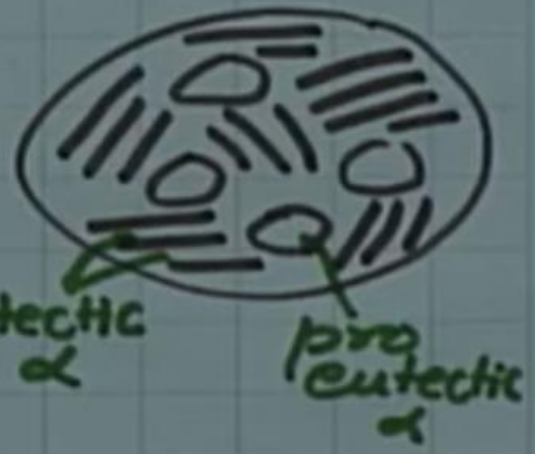
To determine total α and total β in the alloy we draw a the tie-line just below T_E

$$f_{total\,\alpha} = \frac{97-40}{97-18} = \frac{57}{79} = 0.72$$

$$f_{total\,\beta} = 1 - 0.72 = 0.28$$

$$f_{eutectic\,\alpha} = f_{total\,\alpha} - f_{proeutectic\,\alpha}$$

$$= 0.72 - 0.5 = 0.22$$



Eutectic, Eutectoid and Peritectic

- Eutectic - liquid transforms to two solid phases

  $L \underset{heat}{\overset{cool}{\rightleftharpoons}} \alpha + \beta$ (For Pb-Sn, 183°C, 61.9 wt% Sn)

- Eutectoid – one solid phase transforms to two other solid phases

  $Solid_1 \leftrightarrow Solid_2 + Solid_3$

  $\gamma \underset{heat}{\overset{cool}{\rightleftharpoons}} \alpha + Fe_3C$ (For Fe-C, 727°C, 0.76 wt% C)

- Peritectic - liquid and one solid phase transform to a 2nd solid phase

  $Solid_1 + Liquid \leftrightarrow Solid_2$

  $\delta + L \underset{heat}{\overset{cool}{\rightleftharpoons}} \varepsilon$ (For Cu-Zn, 598°C, 78.6 wt% Zn)

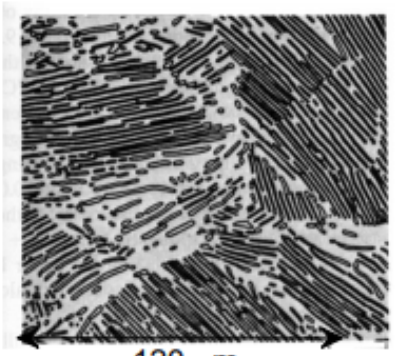
Reference: Prof Grewal's Slide

Iron-Carbon (Fe-C) Phase Diagram

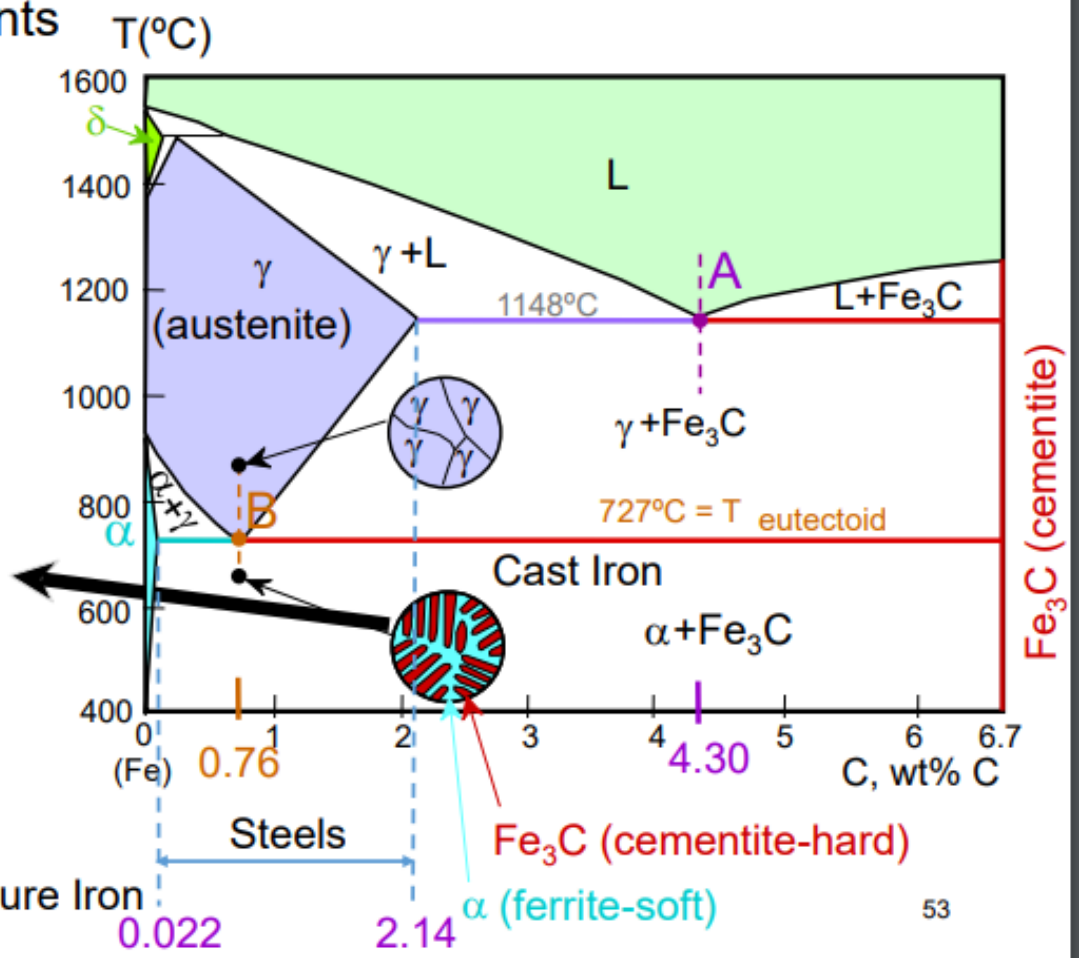
- 2 important points

  - Eutectic (A):
    $L \Rightarrow \gamma + Fe_3C$

  - Eutectoid (B):
    $\gamma \Rightarrow \alpha + Fe_3C$



Result: Pearlite = alternating layers of α and Fe_3C phases



Remember values. Note that x axis doesn't extend till 100%.

Most Imp slide

Solid Phases

α-Ferrite

- Solid solution of carbon in **α** iron
- BCC structure
- Carbon only slightly soluble in the matrix
- Maximum solubility of 0.02% at 727 ℃ to about 0.008% at room temperature

Austenite (γ)

- Solid solution of carbon in γ-iron
- FCC structure; can accommodate more carbon than ferrite; maximum of 2.14% at 1148 C, decrease to 0.76% C at 727 C
- Difference in C solid solubility between **γ** and **α** Is the basis for hardening of most steels.

δ-Ferrite

- Solid solution of carbon in **δ** -iron
- BCC crystal structure
- Maximum solubility of ferrite being 0.09%C at 1495 C

Cementite (Fe_3C)

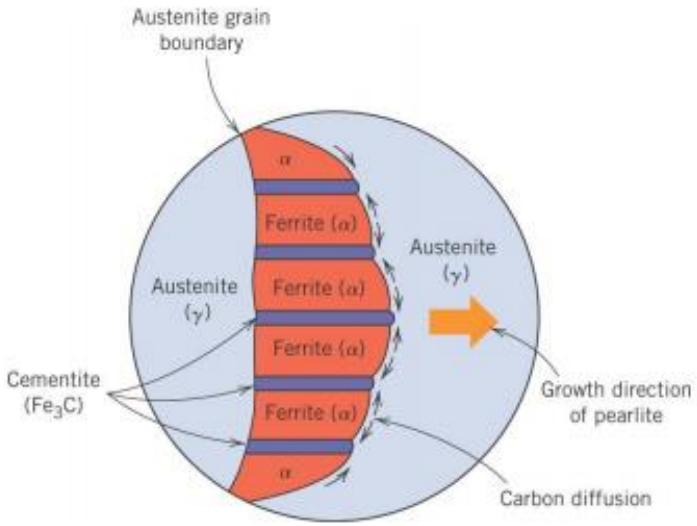
- Intermetallic Fe-C compound
- Orthorhombic crystal structure: hard and brittle

Most Imp reaction in Fe-Fe3C diagram is eutectoid reaction

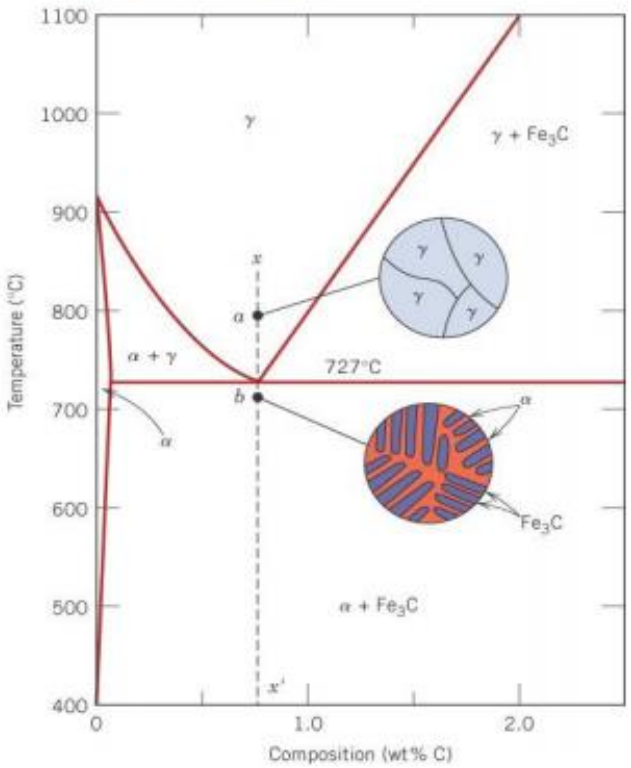
Eutectoid reaction:

$$\gamma \leftrightarrow \alpha + Fe_3C$$

- formation of the pearlite structure
  - nucleating at γ grain boundaries
  - growth by diffusion of C to achieve the compositions of α and Fe_3C (with structural changes)
  - α lamellae much thicker



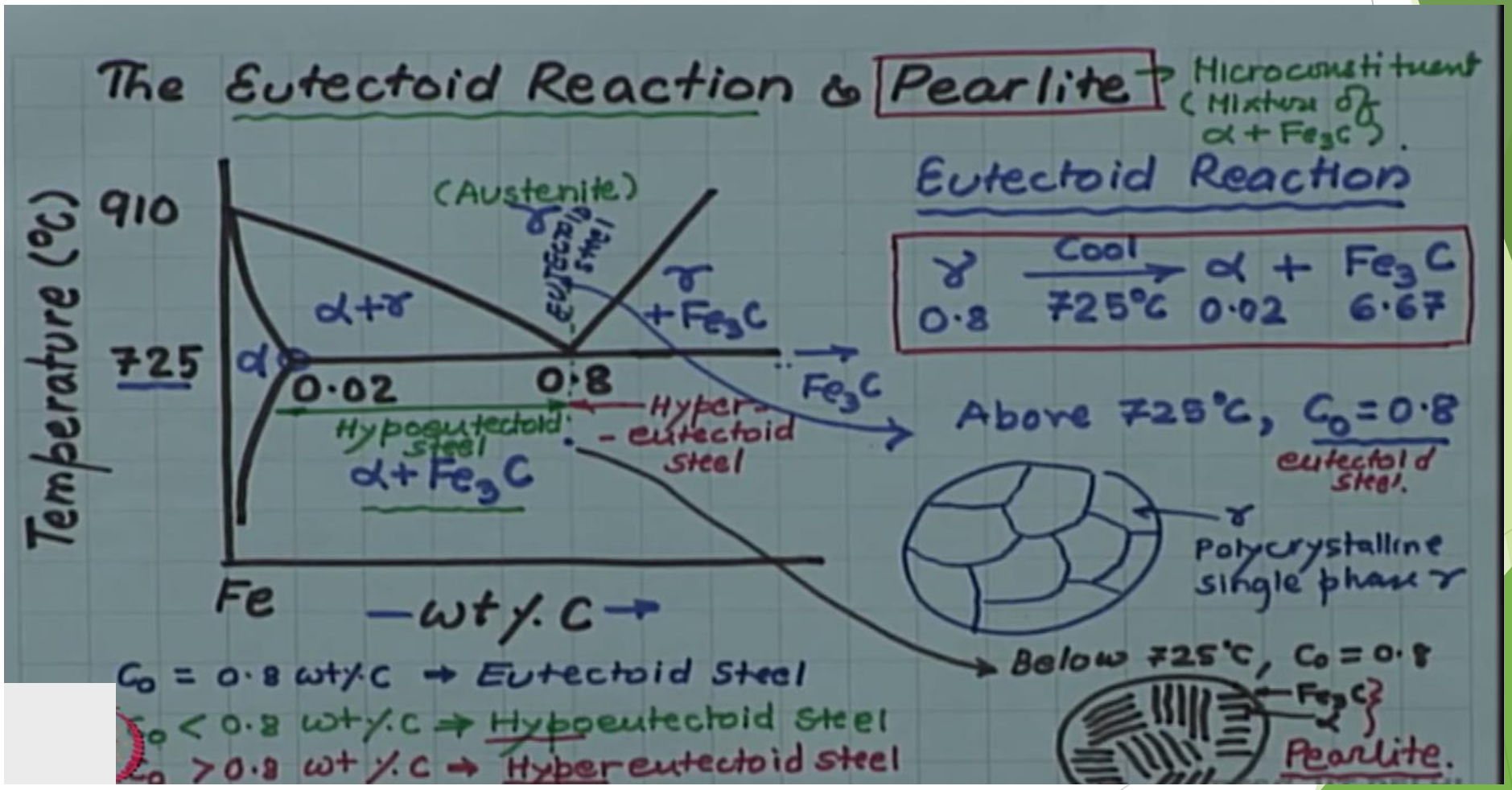
Pearlite



Redistribution of carbon by diffusion

Austenite – 0.76 wt% C
Ferrite - 0.022 wt% C
Cementite - 6.70 wt% C

Eutectoid, Hypoeutectoid and Hypereutectoid steels



Last numerical in Sir’s slide. Very Imp (will be discussed in starting of class 7)

- Concept of hypo and hypereutectoid steel explained there.